

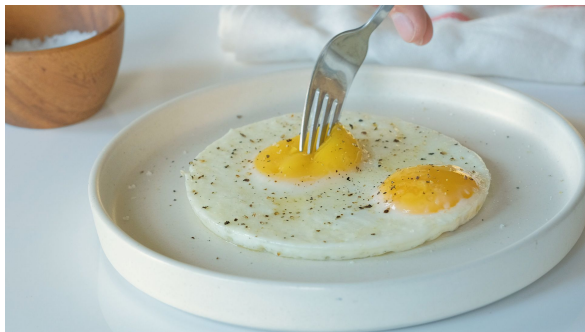


CS 6120

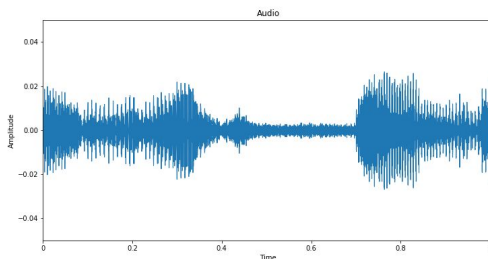
Week 1: Introduction to Natural Language Processing

<http://course.ccs.neu.edu/cs6120f26>

- **Natural Language** - the most compact and efficient way to convey an idea to one another. Distills the salient information, discarding the irrelevant ones. Leverages our knowledge in order to provide full comprehension.



2400000 Bytes
(Compressed, JPG)



50000 Bytes
(Compressed, MP3)

“Two eggs on a plate”

18 Bytes
(Uncompressed)

- **Processing** - in order to effect a desired outcome *for a specific purpose*

Global LLM Integration Footprint (2026)



Global Footprint:

As of 2026, **67% of organizations worldwide** have integrated large language models into core workflows to back their daily operations with generative AI.



Sector Dominance:

Retail and e-commerce lead the commercial LLM landscape, controlling the largest market segment with a **27.5% share**.



Workplace Value:

Professional sentiment is overwhelmingly positive, with **88% of knowledge workers** reporting that LLMs have visibly improved the overall quality of output.



Market Investment:

Capital injection continues to accelerate drastically, as global spending on generative AI technologies reached **\$644 billion by the end of 2025**.



Enterprise **Model Switching** Patterns

66%

Upgraded within Vendor: The vast majority of enterprise customers scale their usage by moving to higher-tier models or enterprise agreements with their existing LLM provider.

23%

Made No Changes:

Roughly one-quarter of organizations maintain their baseline deployment without altering current contracts or models.

11%

Switched Vendors:

A small minority of enterprises actively jump to competing model providers.

Top Switching Triggers

61%

Performance

36%

Cost of Ownership

22%

Safety, Compliance,
Security

9%

Vendor Consolidation

7%

Open-source



Natural Language Processing

Section 0: A brief introduction to the course

- Welcome and hello!
- Applications and industries

Section 1: Administrative and logistics

Section 2: A lab to get you started

Section 3: Some historical perspectives

What is NLP? Why study language and automate it?



Computer programs that analyze, understand and generate (in)formal human language

- The most direct and compact representation of information, intended for communication and conceptual understanding
- Typically translate all forms of modality into linguistic constructs



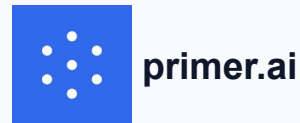
What can NLP do for you? (Before 2019)

Core Capabilities & Applications

Why are you taking this course?

1. Question & Answer
2. Machine Translation
3. Content Understanding
4. Text Summarization
5. Linguistic Analysis
6. Note Taking
7. Fraud and Cheating Detection
8. Topic Discovery

Industry Pioneers & Early Adopters



Quizlet



What can NLP do for you? (Now: 2026)

Core Capabilities & Applications

Why are you taking this course?

1. Moving from "Search and Rules" to Generative AI & Copilots
2. Transitioning from Data Aggregation to Workflow Automation (Agentic AI)
3. Deep Integration with Frontier Cloud Infrastructure (Acquisitions & Partnerships)

Industry Pioneers & Early Adopters



primer.ai



Quizlet



simplr

 innovaccor

1. Healthcare & Clinical Operations

The Business Challenge

Extracting actionable insights from messy, unstructured electronic health records (EHRs) and doctor's notes.

NLP Solution

Automated clinical documentation, medical coding extraction, and patient chart summarization (e.g., platforms like **Innovacer**).

2. Education & EdTech Delivery

The Business Challenge

Scaling personalized student support and tutoring without massively expanding human educator headcount.

NLP Solution

Conversational Socratic AI tutors and automated content explanation (e.g., **Quizlet's Q-Chat** or AI-driven study guide summaries).

3. Customer Experience & Automation

The Business Challenge

Handling massive volumes of Tier-1 customer inquiries quickly and cost-effectively.

NLP Solution

Intent recognition, automated routing, and conversational support agents that resolve queries or summarize call transcripts for human agents.

4. Legal & Compliance Document Processing

The Business Challenge

Reviewing thousands of dense contracts, invoices, and regulatory filings for risk or key obligations.

NLP Solution

Intelligent document processing (IDP) that automatically extracts terms, flags anomalies, and accelerates contract review lifecycles.

Ed Platforms & AI Evolution



Course Hero

Core Focus & Value

A massive digital library of **crowdsourced study materials**, including student-uploaded class notes, study guides, and past exams organized by university.

Strategic AI Shift

Shifted toward AI-driven summarization and document explanations, allowing students to instantly analyze uploaded study guides.

Quizlet

Core Focus & Value

Built on user-generated **digital flashcards and matching games**. Highly popular for active recall, vocabulary learning, and memorization.

Strategic AI Shift

Launched "Q-Chat," an AI tutor built in partnership with OpenAI, acting as a conversational Socratic study coach.

Brainly

Core Focus & Value

A **peer-to-peer social learning network** where students post homework questions and earn points by answering peer questions.

Strategic AI Shift

Integrated AI features to automatically format, clarify, and simplify peer answers, reducing reliance purely on community input.

Bartleby

Core Focus & Value

Owned by Barnes & Noble Education, offering textbook solutions and Q&A as a **direct clone of Chegg Study**.

Strategic AI Shift

Heavily impacted by structural Chegg-like declines; scaled back expansion to protect margins while adapting to AI.

What can you do for NLP?



FUNDAMENTAL RESEARCH

- Fundamental research traditionally led by Universities due to openness, collaboration, and a focus on the long term (10+ years).

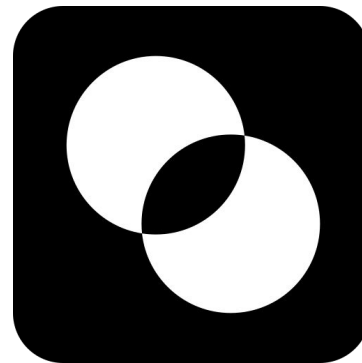
INDUSTRIAL INNOVATION

- Industry contributing significant & ground-breaking advances.
- Changing philosophies to open source, collaboration through funding, shorter horizons.



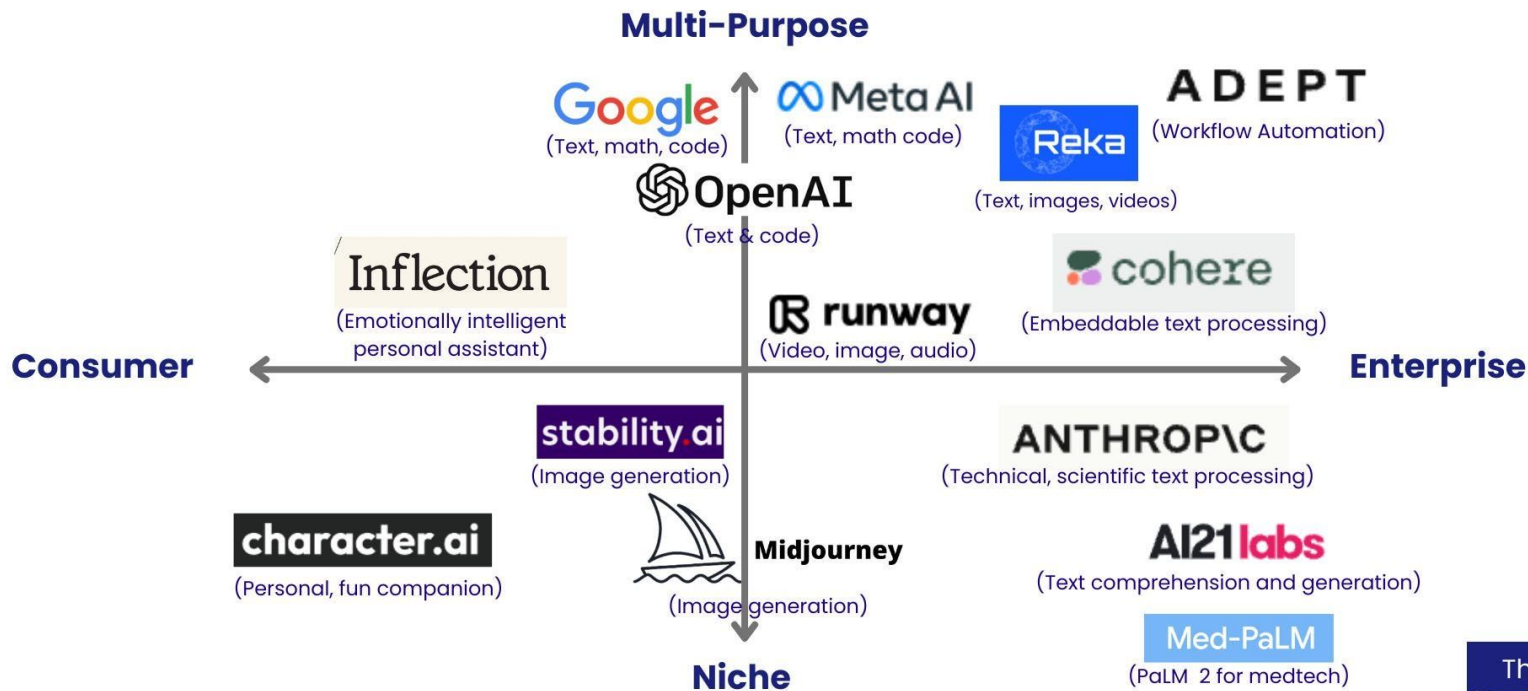


Center for
Research on
Foundation
Models

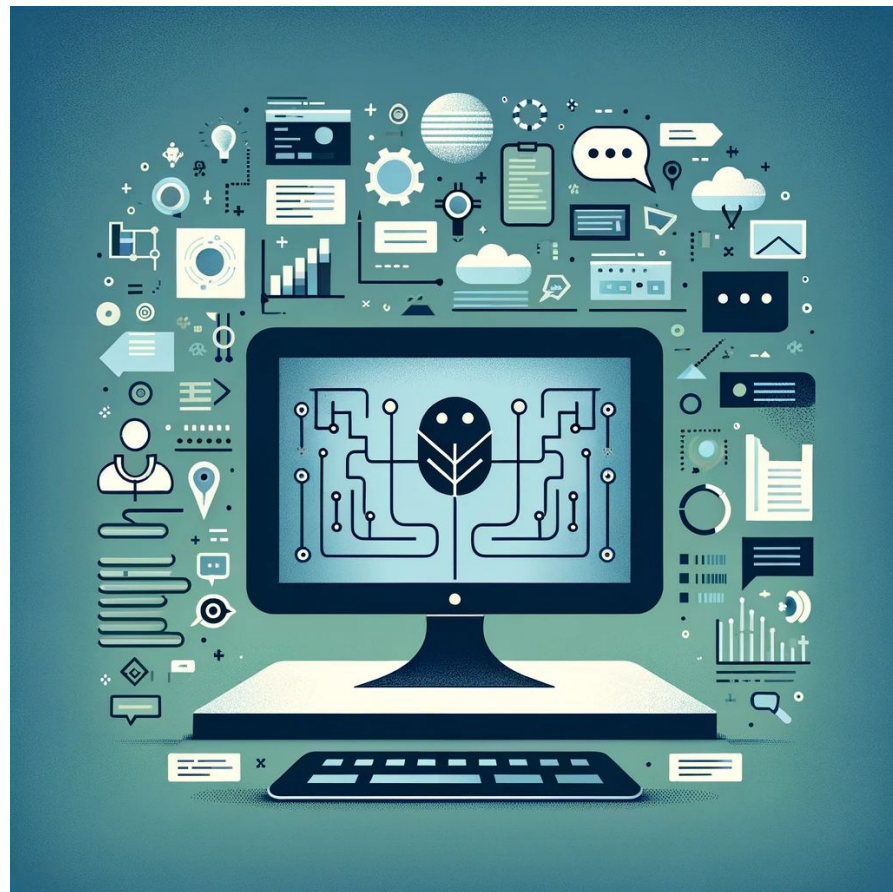


- **Stanford CRFM Foundation Model** Transparency Index: Evaluates major global foundation model developers on governance and disclosure standards.
- **The Global AI Brief** (Top 50 Foundation Models): Publishes recurring rankings of top-performing frontier models.
- **TheirStack / Industry Databases**: Exportable enterprise databases tracking companies deploying foundation models.

Gen AI Companies with Foundational Models



- What is your focus?
- Computational Linguistics
- Recommendation Sciences
- Information Retrieval
- Data Mining
- Applied Machine Learning
- Applied Research





Computational linguistics (CL) is what powers anything in a machine or device that has to do with language—speaking, writing, reading, and listening. It is often linked with natural language processing (NLP), which is a subset of CL.

The usual tech giants: **Google** (including the **NLP research group**), **Microsoft** (including the **NLP research group** in Redmond), **Verizon Media**, **Apple**, etc.

Alelo, **Appen Butler Hill**, **Chenope**, **Decooda**, **Expert System**, **Facebook**, **Intel**, **Lingsoft**, **Lionbridge**, **Mitre, Corporation**, **MultiLingual**, **North Side**, **Nuance**, **Oracle**, **SDL**, **SRI International**, **Systran**, **Vantage Linguistics**, **Verilogue**, **Voiceweb**, **VoxGen**

Modern Marketability of Computational Linguistics & NLP



Modern Definition: Computational Linguistics & NLP power how machines understand, generate, and reason with human language—shifting from handcrafted linguistic rules to scalable foundation models and reasoning systems.

The Legacy Market (Pre-2019)

Core Focus

Rule-based grammars, manual parsing, syntax tagging, and traditional localization.

Primary Employers

Translation bureaus, localization vendors, custom search indexing tools, and legacy enterprise software firms.

Value Prop

Helping computers process structured text via human-engineered rules.

The Generative AI Market (Today)

Core Focus

Frontier model architecture, tokenization mechanics, alignment, safety, and RAG optimization.

Primary Employers

Frontier labs ([OpenAI](#), [Anthropic](#), [Google DeepMind](#), [Meta AI](#), [xAI](#), [DeepSeek](#)) and enterprise AI teams.

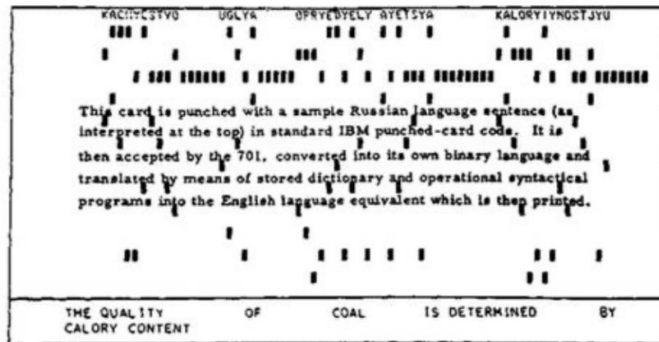
Value Prop

Powering reasoning engines, scaling multimodal systems, and ensuring safe execution.

On the topic of chatbots



How it started

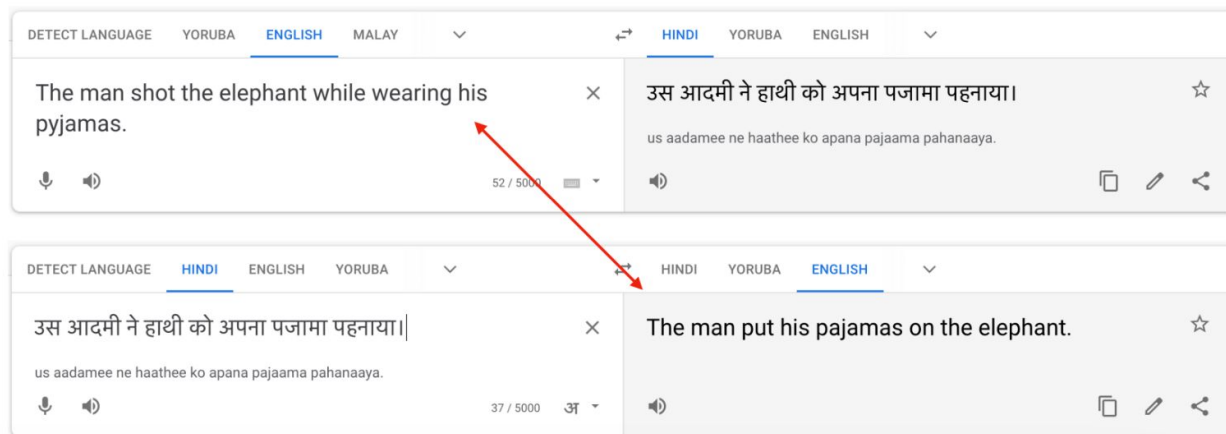


Specimen punched card and below a strip with translation, printed within a few seconds

Georgetown experiment 1954

“Within three or five years, machine translation will be a solved problem”

How it's going



Why is this so hard?



- Ambiguous
- Dialects
- Accents
- Listener
- Humor, sarcasm, irony
- Context, dependencies



**ACCORDING TO THE THESAURUS,
"THEY'RE HUMID, PREPOSSESSING
HOMOSAPIENS WITH FULL SIZED AORTIC
PUMPS" MEANS "THEY'RE WARM, NICE
PEOPLE WITH BIG HEARTS."**



Natural Language Processing

Section 0: A brief introduction to the course

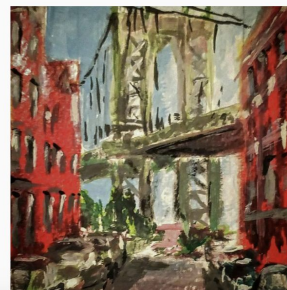
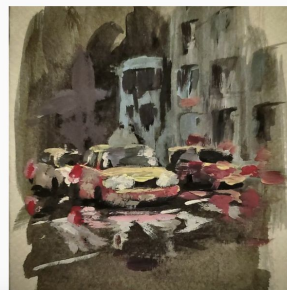
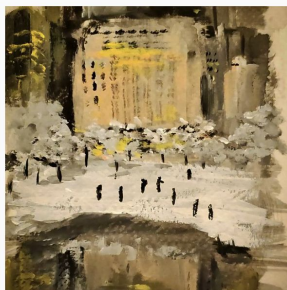
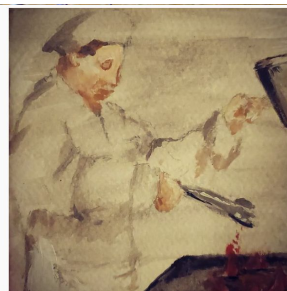
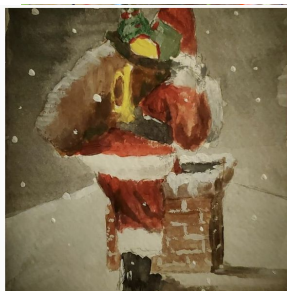
Section 1: Administrative and logistics

- Course objectives
- Syllabus and what you'll learn
- Keynote reading as part of your grading
- Grading, homeworks, exams
- Your open project and groups

Section 2: A lab to get you started

Section 3: Some historical perspectives

- Karl Ni. Etsy, Google, LL / MIT 2009, UCSD 2008, UC Berkeley 2002



- How about you? Align? Industry? PT/FT Students? Taken any ML?

- I did my undergrad in Data Science & AI back in India, then spent some time as ML Engineer @ OSI Digital. I was working on traditional machine learning models and wiring them into OCR & chatbots. Now I'm here doing my MSCS @ Northeastern.
- When I'm not doing CS stuff, I'm usually catching up on basketball and tennis or watching a movie. I'm also checking out new food spots.
- Feel free to reach out whenever you're stuck on anything.





- **Undergrad:** Computer Science, University of Illinois Urbana-Champaign
- **Graduate:** MSAI, Northeastern University
- **Professional Experience:** Ex-WeRide.ai (Autonomous Driving)
- **Interests:** Classical Guitar, Badminton, Go





- What are the courses that you have taken?

Course Title	NEU Course Number
Foundations of AI	CS 5100
Machine Learning	CS 6140
Generative AI	CS 6180
Advanced Machine Learning	CS 7140
Deep Learning	CS 7150
Advanced Algorithms	CS 5800 / CS7800



- Course objectives
- Syllabus and administrative
- Keynote reading
- Your open project and groups



Our class website (CS6120) is at:

- <https://course.ccs.neu.edu/cs6120f26> Accessible via Canvas
- You can find our syllabus, reading, homeworks, project templates, etc. there.

Primary form of communication will be [Teams](#)

- (Ignore any new Teams created by Khoury.)

Primary grade submission and shareout will be [Gradescope.com](#)

Objectives of the course



By the end of this course, you will build competencies in your:



knowledge-base



implementation fluency



industry/academic skill

By the end of this course, you will have ...



... built your foundational skills

You'll have

- intuition with first-hand experience of the inner workings of objectives, the building blocks of DNN architectures, and gradients in backpropagation
- literacy to understand over 80% of the modern papers in the area of natural language processing
- the ability to teach yourself new skills



By the end of this course, you will have ...



... improved your fluency and practical knowledge



You'll be able to code with velocity by

- leveraging the latest toolboxes, e.g., PyTorch and Tensorflow or common Python ones
- being able to debug NLP algorithms and recognize pitfalls and identify when things go wrong
- understanding where algorithms are useful with real-world practical applications

By the end of this course, you would have ...



... built your accomplishments and resume

Your track record will have a marketable product that can be scaled to real-traffic and with common PROD instruments

You will do this by:



knowledge-base



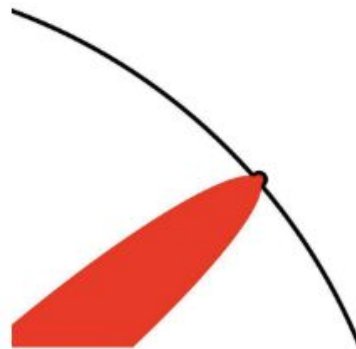
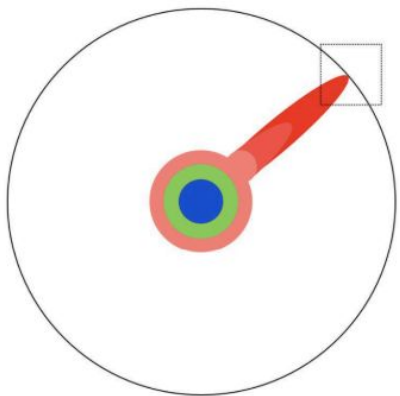
implementation
fluency



industry/academic
skill

- Keynote Papers - Build literacy and the ability to self-teach
- Homeworks and Labs - Practice with implementation and debugging
- Project - Build your real-world experience and track record

- This class is an elective (wonderful to see the interest!)
- The field is rapidly advancing
 - Class is more heavily paper driven
 - Recent advances being leveraged in practice ***are not in books***



ML is an advanced class: this is a *very* advanced class

Recitation Survey



Week	Class Topic	HW Due	Recitation Topic
1	Introduction & Applications of NLP		Intro to Python
2	ML Foundations and Software Engineering	Autocorrect	Linear Algebra
3	Language Classification / Naïve Bayes		Multivariate Calc
4	Autocorrect - Text Processing Algorithms		Intro to TF / Keras
5	(No Lecture)	Text Classification	
6	Introduction to Language Modeling (N-Grams)		
7		Min Edit Distance	

Any other course topic?



Structured towards *industry practice* of using Natural Language Processing

- Productionizing NLP (Lec 1,2,9,10,11 and Lab 1,2,7,11,11)
- ~~Named entity recognition~~ (Lec 7)
- ~~Sentiment analysis~~ (Lec 3, 6)
- ~~Group documents by topics~~ (Lec 7)
- Retrieving information from corpora (Lec 7)
- ~~Machine Translation~~ (Lec 7)
- Text Summarization, e.g., for Reviews / Paragraphs (Lec 8,9)
- (Chatbots) Following instructions, e.g., How do you code with LLMs? (Lec 10, 11)
- (Chatbots) Ensuring compliance with your use case, e.g., “Tell me how to hack into neighbors WiFi?” (Lec 11)
- Representations of text (Lec 4, 6)
- ~~Salient parts of text~~ (Lec. 3, 4)
- Text classification (Lec 2, 3)
- LLMs in and for practice (Lec. 10,11)



Structured towards *industry practice* of using Natural Language Processing

Lectures build towards modern Large Language Modeling (LLM)

- The origins and theory of modeling language
- (Pre)processing language in practice
- Light approaches (non-DNN) to modeling text
- Neural Network fundamental building blocks (e.g., embeddings & attention)
- Heavy towards leveraging & gearing LLMs towards practice
 - Summarization (e.g., for Reviews / Paragraphs)
 - Following instructions (e.g., How do you code with LLMs?)
 - Ensuring compliance with your use case (e.g., “Tell me how to hack into neighbors WiFi?”)
 - Using them for information retrieval / reading text for you

What Lectures We Won't Be Covering (And Why)



- [Last Semester's Lectures](#)
- **David Blei's Latent Dirichlet Allocation (LDA)**
- **Combine HMM and RNN Lectures (Sequential Models)**



- Course objectives
- Syllabus and administrative
- Keynote reading
- Your open project and groups



You will do well in this class if:

- You do the relevant reading and come to class with questions about it
- Start on the homework early and come to TA office hours about any Q's
- Think about the practical engineering aspects of the lab
- Finish your project to create a single NLP delivery

Some suggestions to do excellent in the class and beyond:

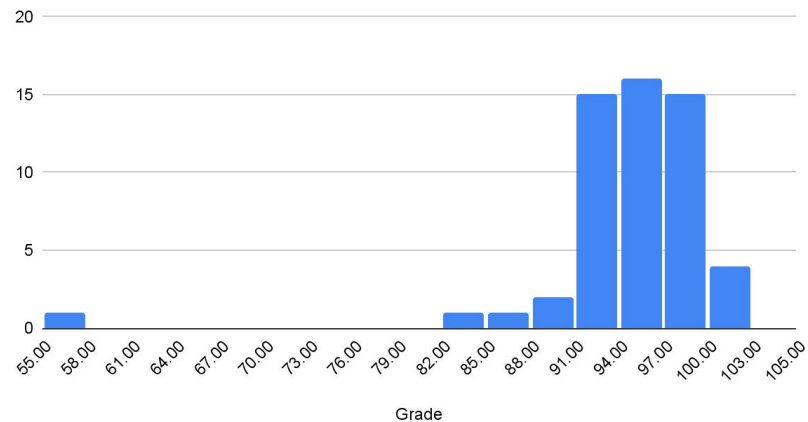
- Keep up with technology in the AI World
- Replicate papers in the homeworks and during TA office hours
- Start on homeworks early and preview the lectures

It's a bit of work, but ... it's worth it!



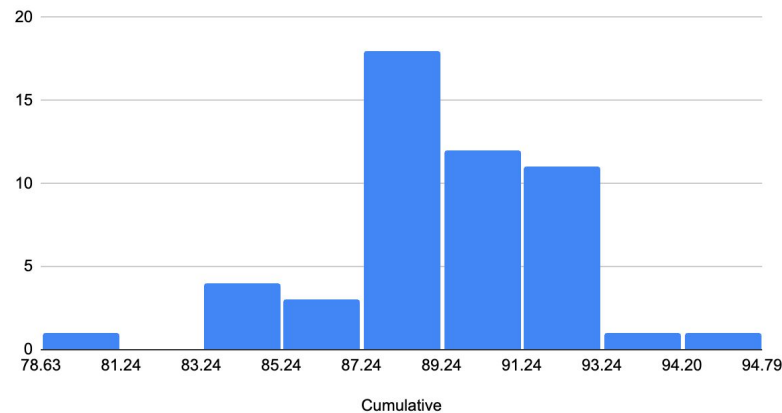
Curved to A (Pretty much everyone got an A or A-): In the Clump

Histogram of Grade



Spring 2025

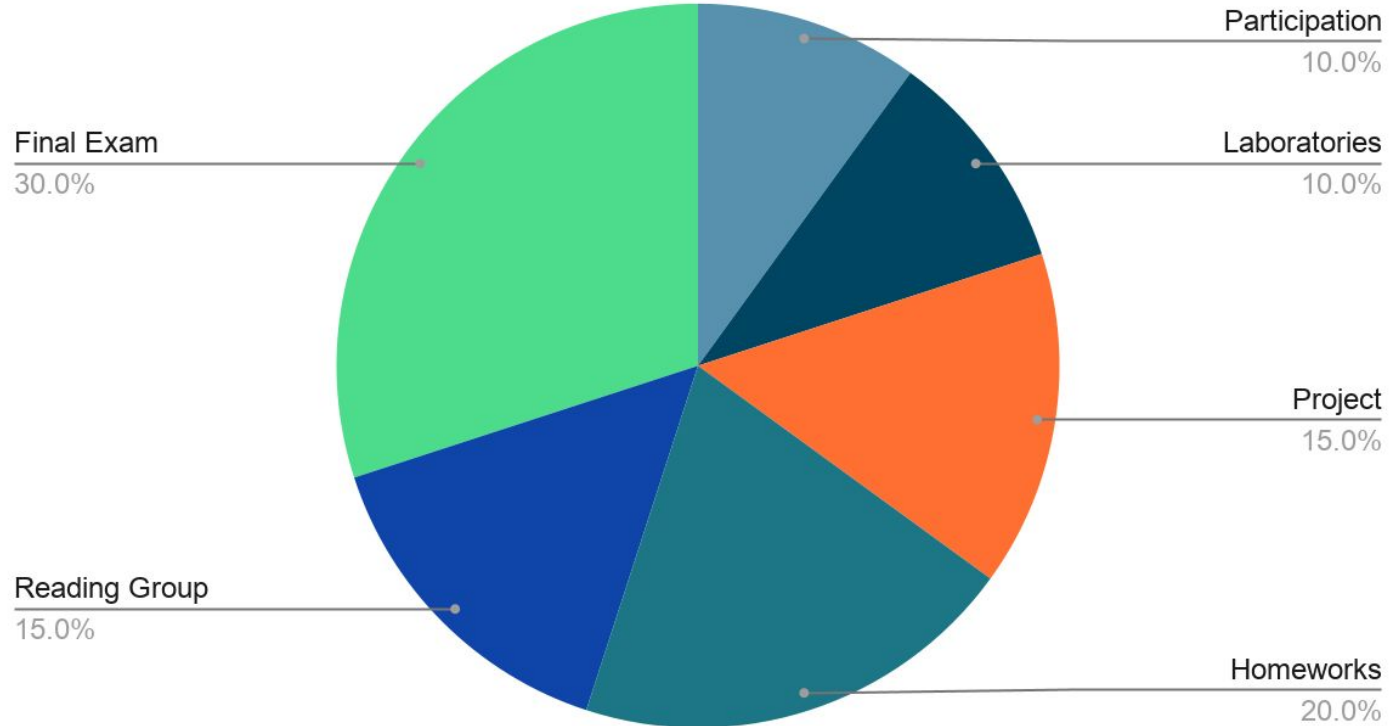
Histogram of Grades



Fall 2025



CS6120 Scoring





(Tracked Actions)

- At the beginning of each class, 5 minute time for “this week in AI / LLM”
 - Keep track of your participation, this week in AI/ML
 - Come up with at least one interesting news tidbit during the semester
- Contributions to and presence in group activities (reading, project, etc.)
 - Participation and getting along with your fellow students!
- Weekly attendance and lab submissions
 - To avoid absences, please e-mail staff 24 hours in advance

(Untracked / Subjective Actions)

- Discussions on paper videos during class
 - Shows you’ve also read the papers
- Questions in class on lecture
 - Lecturing vs Learning Session - Questions and Answers
 - I don’t care if you’re wrong!



- Labs in class: completed before the next lecture (time in class devoted)
- Labs: applications for you to understand and try
 - Pass / Fail - Effort Based
 - Correct solutions unnecessary.
 - GCP / GPU's aren't always available.
- Lab assignments need to be handed in the next week (following assignment)



- Suggested Reading:

- [Speech and Language Processing](#), Jurafsky & Martin

Open source, data proliferation and compute $\Rightarrow$
this field moves faster than textbooks can keep up

- Required Keynote Reading:

- Will be math heavy, but well-resourced
- Not expected to know the math off the bat
- By the end of the semester, you should!



- Seven Assignments

	Topic	Assigned	Due
Assignment 1	Intro to the Class	Today!	September 21
Assignment 2	Deep Learning	September 26	October 5
Assignment 3	Autocorrection	October 10	October 19
Assignment 4	Intro to Language Models	October 24	October 26
Assignment 5	Word Representation	November 7	November 2
Assignment 6	Sequence Models	November 14	November 9
Assignment 7	Transformers	November 21	November 16

A Single Exam

- Historical Context
- Will be closed book
- Allowed 1 sheet of paper
- No devices or electronics allowed
- Very similar to your homework.
- Everything will be provided
- Six quizzes available from last semester
- Tests the application.

ECON 1170 Midterm & Final Scores

After Serrano made the final in-person, 18 students dropped the class and nine students did not take the exam. The scores of the remaining 59 students are displayed here.





- Reading group videos: ***linked the date on the sheet*** (previously week after)
- Laboratories: ***cannot be handed in late***
- Assignments: can be up to two weeks late with penalties
 - 3 points each work day until 15 points.
 - Maximum deduction is 15 points
 - Final submission deadline the following week
- ***Quizzes must be taken in class***
 - They will not be excused without prior rationale

Typical Class Agenda



Timing	Topic	Notes
5-10m	Administration	
5m	This Week in ML	
5m	Paper Review / Videos	
30m	Laboratory	Sometimes More
2h	Lecture	Broken Up Over 3 Hours



- Course objectives
- Syllabus and administrative
- Keynote reading
- Your open project and groups

What *are* required “keynote papers”



- Papers that are core curricula to this course are “keynote papers”
 - They are the most widely read and cited in the community
 - They are papers that are considered the fundamental elements of NLP
 - Comprehensively, the papers will make you fluent to read most others
- Signup Sheet Available Here
 - **Later weeks may often be *easier* than earlier weeks.** Difficulty is not ordered.

Use your resources! There are so many of them



- Difficulty in Reading

- Several attached resources, but these aren't comprehensive
- Because they're fundamental elements, they are very well resourced
 - Use YouTube videos, blogs, and ChatGPT for understanding



Other things you can do to better understand the paper



- Replicate the results (help in TA office hours)
- Explain it to yourself (maybe record it and make a vlog?)
- Summarize it in a blog post (there are so many examples)
- Discuss it with a friend before or after class



- **Video (5 min) → Posted Online**

- **Summary Video:** A layman's description in easy-to-understand language with examples and real-world scenarios.
- **Elevator Pitch:** To the classroom, pitch the main ideas of the papers, what makes them advantageous, how they are relevant to your daily lives and can make them better. Think of these as VC pitches.

- **Document (1 page) → Posted Online**

- **Summary Text:** This should your interpretation to an easy-to-understand summarization of the paper. It should reflect your perspectives and opinions and closely track the video.

- **Q/A & Defense: (5 min) → Optional In-Class**

- **In the Classroom:** defend this paper while acknowledging its shortcomings and challenges.



WEEK 6 OCTOBER 26

Word Modeling with Self-Supervision

Perhaps the most influential paper to have come out of the natural language community is the word2vec paper. This week's lecture reviews word models (including word2vec as well as continuous bags of words) and the embeddings / representations that they create.

LECTURING TOPICS

- Embeddings with Continuous Bag of Words (CBOW)
- Skip-gram and Negative Sampling
- Intrinsic and Extrinsic Evaluation
- From Words to Sentences

SUBMISSIONS



LABORATORY ASSIGNED

Laboratory - Word Embeddings with CBOW



LABORATORY ASSIGNED

Laboratory - Mikolov's Word2Vec (Optional)



HOMEWORK ASSIGNED

Assignment 5: Word Embeddings

LITERATURE & REFERENCE



READING GROUP

Learning Text Similarity with Siamese RNNs

Suggested keynote reading

Videos for keynote paper

Blogs for keynote paper

Topic in the week's related to keynote paper

Sign Up, Assemble, and Link



- [Signup sheet \(under readings\)](#) - on the [class website](#)

Keynote Reading Schedule

Keynote papers are required reading that underly the fundamentals of modern natural language processing theory and practice. Each week, students should sign up for a paper. [Please do so here](#). Updates (it might take a few minutes) on the current schedule can be viewed here:

Keynote Signup : Fall 2025			
	Paper	Name	Summary, Elev.
3	9/22 GPT-5 Model Card (OpenAI)	Name	Name
4	9/29 Outrageously Large Neural Networks: The Sparsely-Gated Mixture of Experts	Name	Name
5	10/6 Model Context Protocol (MCP): Landscape, Security Threats, and Mitigations	Name	Name
7	10/20 Topic Modeling - Latent Dirichlet Allocation (Blei et al)	Name	Name
8	10/27 Distributed Representation of Words and Phrases and their Compositions	Name	Name
9	11/3 A Survey of LLMs Including ChatGPT and GPT-4	Name	Name
9	11/3 Learning Text Similarity with Siamese Recurrent Networks	Name	Name
10	11/10 Long Short-Term Memory (Hochreiter & Schmidhuber)	Name	Name
10	11/10 On the Difficulty of Training RNNs (Pascanu et al)	Name	Name
11	11/17 Attention is All You Need (Vaswani et al)	Name	Name
11	11/17 BERT - Pre-training Bidirectional Transformers	Name	Name
12	11/24 Training Language Models to Follow Instructions with Human Feedback	Name	Name

Fall 2025

Link to the paper

Fill out your name and link to your work

- Link to the video (YouTube, Vimeo, Google Drive, or OneDrive)

Video (5 Minutes) and Summary (1 Page)



- Brief Overview of the Work
 - Key equations and interests
- Elevator Pitch for the Methodology
 - Why was this important?
 - How do you think it can be leveraged?
- Challenges of the Approach
- Potential Applications
 - Be creative!
 - Think about many different verticals



1 Pager

EMBRACE THE WORLD!
CELEBRATING CULTURAL DIVERSITY AND GLOBAL CONNECTIONS

- **Exploring Cross-Cultural Communication:** Understanding the nuances of different languages, customs, and traditions to foster meaningful connections and bridge cultural gaps.
- **Global Citizenship:** Nurturing a sense of shared responsibility and empathy towards diverse cultures, promoting inclusivity, and fostering a harmonious coexistence in our interconnected world.
- **The Power of Cultural Exchange:** Embracing internationality as an opportunity for personal growth, expanding horizons, and gaining new perspectives through engaging with people from diverse backgrounds.
- **Cultural Sensitivity in a Globalized World:** Developing an awareness of cultural norms, values, and practices to navigate intercultural interactions with respect, appreciation, and adaptability.
- **Promoting Unity Through Diversity:** Recognizing the strength that comes from embracing multiculturalism, fostering tolerance, and fostering a sense of global community while celebrating our unique differences.

GLOBAL EDUCATION FOR TOMORROW'S LEADERS!
EQUIPPING STUDENTS WITH THE KNOWLEDGE, SKILLS, AND INTERCULTURAL COMPETENCIES NECESSARY TO THRIVE IN AN INCREASINGLY INTERCONNECTED AND MULTICULTURAL WORLD.

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PROSHRED
SECURITY FIRST

COMPLYING IN THE EDUCATION INDUSTRY
Protect your privacy like you protect your kids!

PROTECT YOUR STUDENTS AND YOUR REPUTATION
State and federal regulations like FERPA require educational facilities to protect the confidential information of their students and instructors. Having a regularly scheduled information destruction service makes your learning center compliant and trustworthy with these regulations in your community.

SHREDDING ON-SITE IS AS EASY AS 1-2-3
Our secure shredding process starts the instant you deposit confidential documents into our locked security conveyance. Based on your service schedule, our customer service professionals transport your sensitive information directly to the shredding truck parked outside your location.

Our on-site process is certified to be the most secure, seamless and convenient method of information destruction and privacy compliance. You can even watch your documents being destroyed through a monitor on the truck.

RECYCLE SHREDDED DOCUMENTS
We all have a stake in preserving our environment, and PROSHRED takes this responsibility seriously. We recycle all the shredded paper that we destroy, partnering with you to increase your information security commitment to recycling for the benefit of the environment.

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NSF AAS

Making Your Videos Entertaining!



In communication - style is important / spend some time on how you say it

[Diana Hsieh](#) - Knowledgeable and Interesting

[Harper Carroll](#) - Confident and Assertive

[Siraj Raval](#) - Energetic, Funny, Interesting.

[Annie Sexton](#) - Multiple Cuts, Insertion of Perspectives

[Two Minute Papers](#) - Modern and High Production

Make Interesting Videos



- Have fun graphics, add sound effects
- Include quotes from industry experts / cross-referenced fun facts
- Applications: How it affects your daily life / where can you see it in action?



Easy ways to create



Educational Videos For YouTube

Start Recording



- Course objectives
- Syllabus and administrative
- Keynote reading
- Your open project and groups

Class Project: Build a RAG System



- Objective: Design, implement, and demonstrate a complete Retrieval-Augmented Generation (RAG) system.
- Data: This is where you have degrees of freedom:
 - Must be large in scale (1k+ articles)
 - Examples: academic papers, news feeds, specific blogs, structured data, shopping catalogues, mathematical texts, etc.
- Project Requirements:
 - The LLM Component ***must be local***.
 - Citations must be in the response



- Finished Product with Front End
 - Supplied: Streamlit templates in class labs
 - Requires:
 - Readme.md, includes how to replicate / run algorithms
 - Dockerized container for platform independent engineering
 - Technical documentation that clearly states principles leveraged from class
 - Code that is ChatGPT-free, run through automated checking
 - Advantage: 20% of grade, and does not require additional work beyond that taught in class
 - Disadvantage: Not a single shot to get an A in the class
- Easily achievable with front end examples in in-class laboratories
- Focus isn't on novelty, but should have some elements of it (e.g., new data, new business use case, etc.)

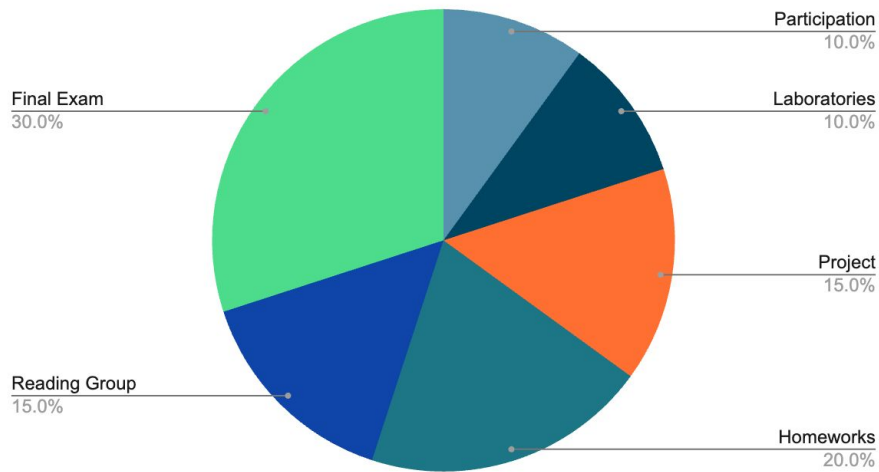
Product Delivery to be Scaled

Fully Packaged NLP Product

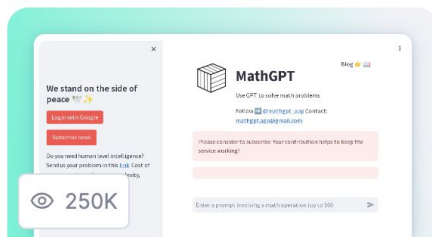
- Works in real-time
- Front and back end
- Can leverage cloud resources
- Containerized and can be run by TA
- Novel in some way (i.e., your data, etc.)



CS6120 Scoring



Example projects with front ends

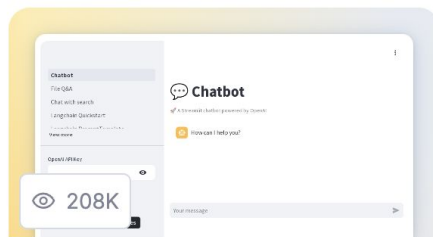


MathGPT



napoles-uach

[View source →](#)



LLM examples



streamlit

[View source →](#)

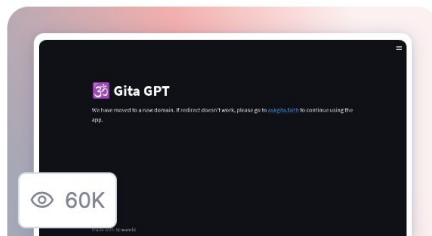


ChatGPT with Memory



leo-usa

[View source →](#)

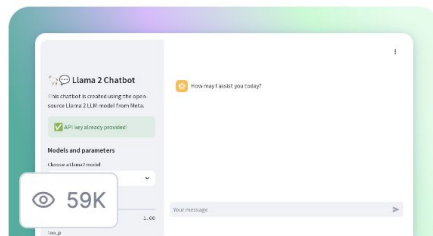


Gita GPT



kinshukk

[View source →](#)

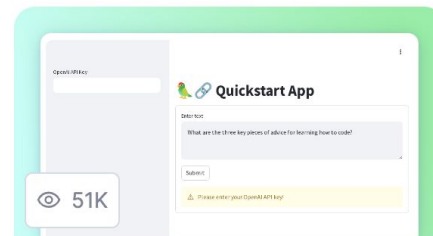


Llama 2 Chatbot



dataprofessor

[View source →](#)



Quickstart App



dataprofessor

[View source →](#)

- Containerized application that can be scaled into production
- Criteria -
 - Must be replicable and containerized
 - Must be functional and operating with expected parameters
 - Must be robust to all inputs
- Deliverables
 - Technical report with README.md, and Github Repository with Docker environment



SPIE Publication - Deadlines (Post-Class)



- Novel and Interesting Projects: Submit to [SPIE Applications of ML 2027](#)
- Semester Ends December 2026
- Abstract Submission Deadline (February / March 2026)
- Paper Submission Deadline (June 2026)

The screenshot shows the SPIE OPTICS+ PHOTONICS website. At the top, there are navigation links: "Attend", "Program", "Exhibition", and "Presenters". The "Program" link is highlighted. Below this, there is a sub-navigation bar with links: "Program Home", "Browse the program", "Conferences" (which is highlighted with a red underline and a dropdown arrow), "Special Events" (with a dropdown arrow), "Awards Banquet", and "Courses". To the right of the navigation bar, the text "3 - 7 August 2025" and "San Diego, California, US" is displayed. Below the navigation bar, the page content shows "Conference 13606" followed by the title "Applications of Machine Learning 2025" in bold. Below the title, the dates "4 - 6 August 2025" and location "Conv. Ctr. Room 8" are listed. At the bottom, there is a button with a calendar icon and the text "Add to My Schedule".

- San Diego - Professional Development Fund for Registration & Travel



- Course Objectives
- Required Keynote Papers
- Project and Its Rubric
- In-Class Labs, Quizzes, and Homework
- **Dropped Lectures**



- [Get the link and try some equations](#)

- Try some equations

```
\begin{equation}
```

```
  X \in \mathbb{R}^{10}
```

```
\end{equation}
```



Natural Language Processing

Section 0: A brief introduction to the course

Section 1: Administrative and logistics

Section 2: A lab to get you started

Section 3: Some historical perspectives



- <https://course.ccs.neu.edu/cs6120f26/assets/pdf/lab-1.pdf>
- Granted \$50 for the Spring; starting 9/1 until 2027

Course Start Date: 9/1/2026

Students' email domain(s): @northeastern.edu, @ccs.neu.edu

Students can request coupons from the URL and redeem them until: 1/1/2027

Coupons Valid Through: 9/1/2027

Number of Coupons: 60

Face Value of Coupon(s): USD 50.00

- **Make sure you close instances & ensure you aren't incurring costs!**

Check how many credits you have



 Start your Free Trial with \$300 in credit. Don't worry—you won't be charged if you run out of credits. [Learn more](#)

 Google Cloud

 CS6220

Search (/) for resources, docs, products, and more

 Search



Billing



Credits

[ALL CREDITS](#)

View and download credit details here. Active committed use discounts are not included here and can be viewed on the [Commitments page](#).

 Filter Filter credits

Credit name	Status 	Percent remaining	Remaining value	Original value	Type	Credit ID	Scope 
Data Mining	 Available	94%	\$94.41	\$100.00	One-time	90UCJ25J...	Any service

 Budgets & alerts

 Billing export

Cost optimization

 FinOps hub

 Committed use discounts...

 CUD analysis

 Pricing

 Cost estimation

 Credits

Billing management

 Account management

 Release Notes



Natural Language Processing

Section 0: A brief introduction to the course

Section 1: Administrative and logistics

Section 2: A lab to get you started

Section 3: Some historical perspectives

Where are we today?



We're currently in one of the longest periods of sustained interest in AI in history because:

- (1) Today's distributed systems dwarf the computing power of the past and
- (2) there are vast troves of training data on which AI systems can cut their teeth

Many doubt AI's ability to pass the Turing Test and prove its ability to create systems that imitate human intelligence and behavior.

It's still an open question how far the technology can go...and how far **you** can push it.

How we got here?



- AI “Summers” and “Winters”: comes down to \$\$
- Long term and fundamental research: typically government
 - Department of Defense grants
 - DARPA provides University grants
 - Has led to the founding of many companies
 - Criticisms are: beltway bandits
- Neural Networks for NLP co-adapted with Neural Networks from other Domains





- **History of the Applications of NLP**
- Timelines of Modern Technological Advances in NLP
- Applied NLP in the Age of LLMs

A Coarse Timeline



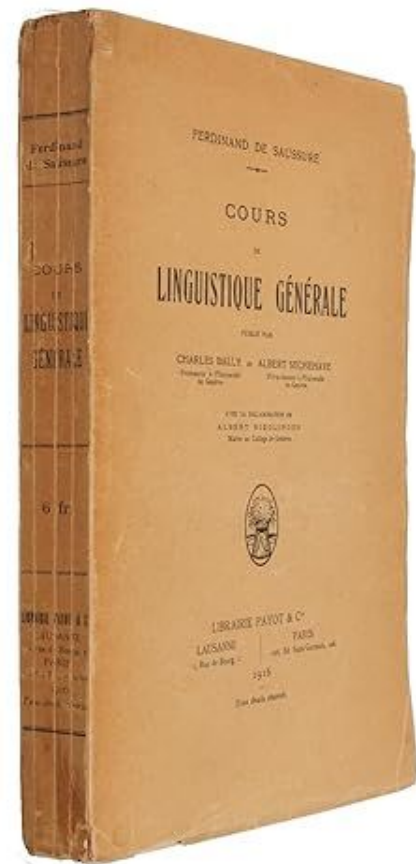
1916	de Saussure develops General Linguistics Course
1950	Turing writes Computing machinery and Intelligence
1954	Georgetown experiments
1966	ELIZA is the first chatbot
1975	The First AI Winter
1980	If/Else computing revives AI research
1987	The Second AI Winter
1990	Statistical methods take the community by storm. SVMs are developed and become popular. VC dimension established
2001	The first neural language model is built
2012	ImageNet makes deep learning the de facto AI poster child
2013	Mikolov writes word2vec and uses skipgrams
2014	Sutskever writes about sequence-sequence models, popularizing RNNs
2015	Attention modeling is introduced
2017	Google creates Transformer neural networks
2019	OpenAI becomes for-profit enterprise
2022	OpenAI releases ChatGPT
2024	LLMs proliferate throughout the world

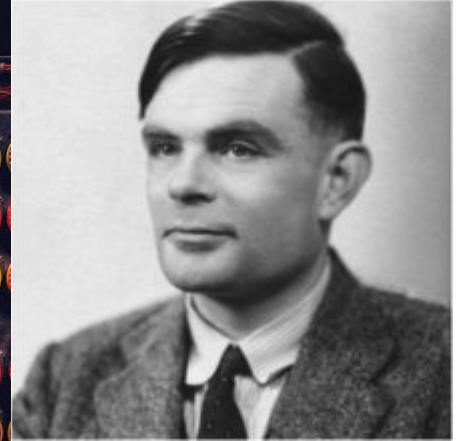
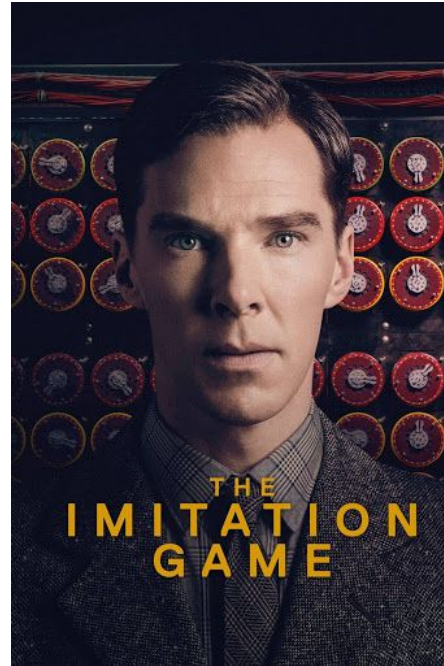
Almost Ending Before It Started (1916)



Course in General Linguistics

- Ferdinand de Saussure - Developed foundation for modeling language as systems (structural linguistics)
- Died in 1913 before publishing work
- Albert Sechehaye and Charles Bally (colleagues) - gathered notes of his students, wrote the book: *Cours de Linguistique Générale*, evolving into NLP.
- Published in 1916



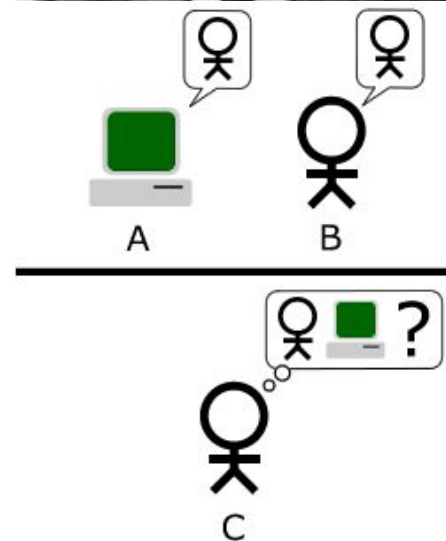


The Turing Test (1950)



The Imitation Game, a.k.a., the Turing Test

- Cryptics and codebreaking during World War II
- Father of modern computing through the ***Turing Machine***
- Wrote 1950 Paper: *Computing machinery and Intelligence*
 - Opening: “I propose to consider the question, ‘can machines think’”
 - Becomes the philosophy of modern artificial intelligence
 - Develops the Turing Test

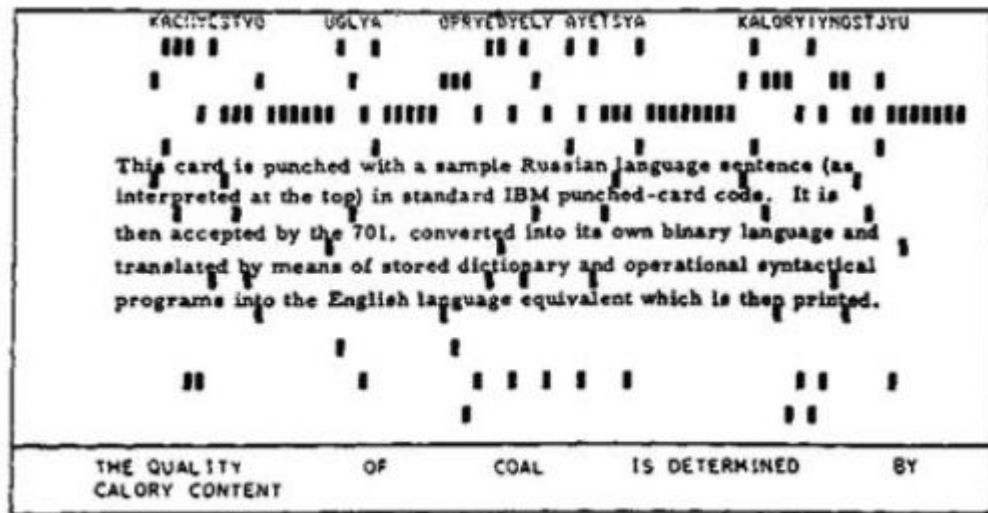


The Georgetown / IBM Experiments (1954)



Purpose: attract governmental and public interest and funding by showing the possibilities of machine translation

“Within three or five years, machine translation will be a solved problem”



Specimen punched card and below a strip with translation, printed within a few seconds

The First Chatbot: ELIZA (1960s)

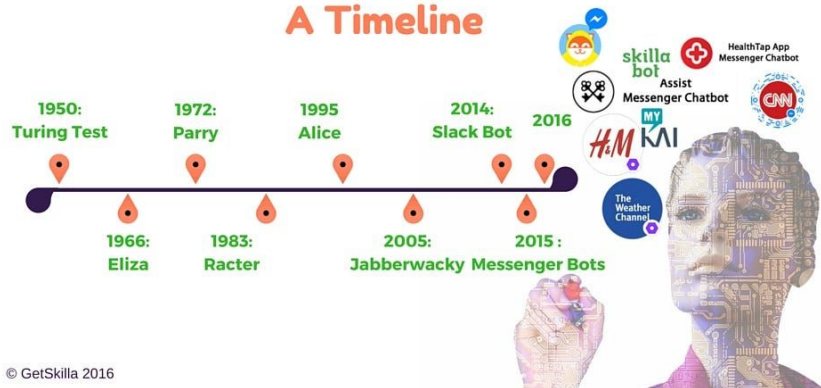


The Rogerian Arguments and Psychology

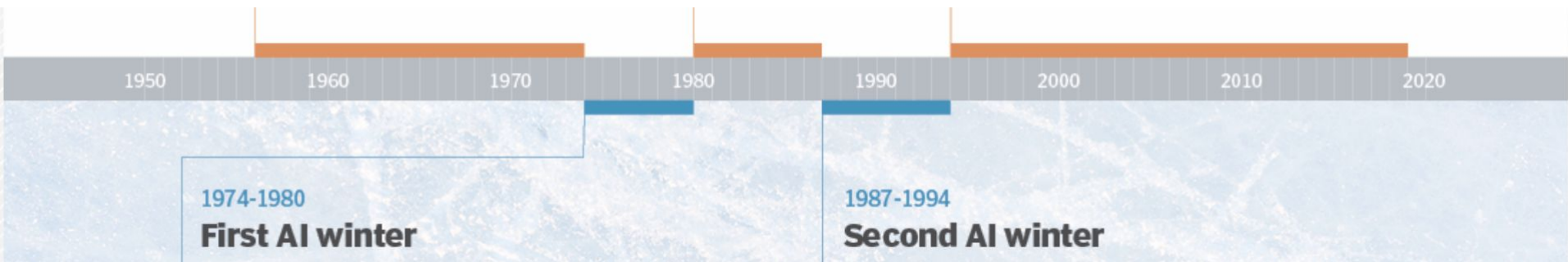
User: I am unhappy.
ELIZA: DO YOU THINK COMING HERE WILL HELP YOU NOT TO BE UNHAPPY
User: I need some help, that much seems certain.
ELIZA: WHAT WOULD IT MEAN TO YOU IF YOU GOT SOME HELP
User: Perhaps I could learn to get along with my mother.
ELIZA: TELL ME MORE ABOUT YOUR FAMILY
User: My mother takes care of me.
ELIZA: WHO ELSE IN YOU FAMILY TAKES CARE OF YOU
User: My father.
ELIZA: YOUR FATHER
User: You are like my father in some ways.

Weizenbaum (1966)

History of Chatbots: A Timeline



The AI Winters



AI Winters - A Chill in the AI Enthusiasm

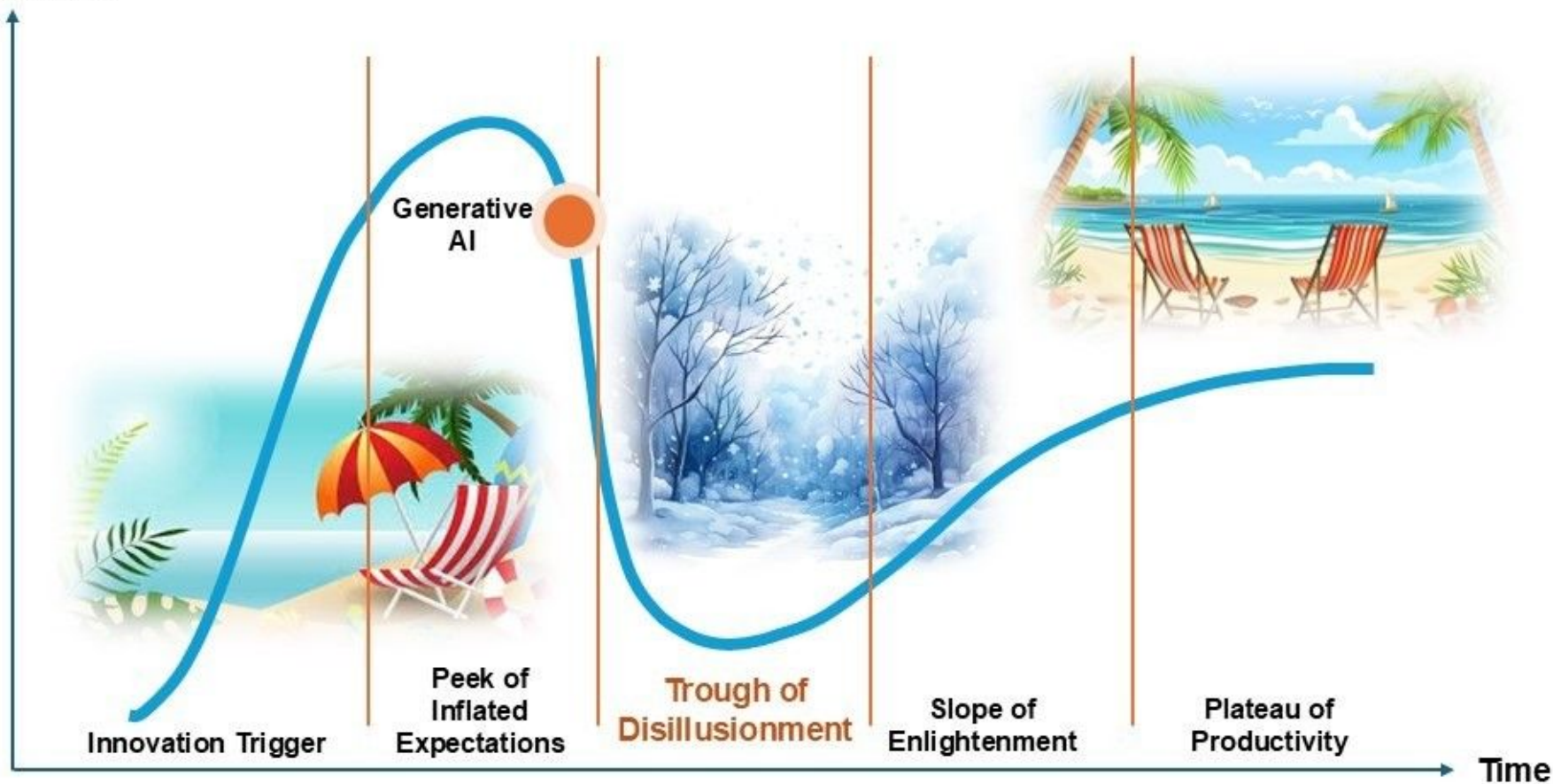
An AI winter refers to a period of reduced funding and interest in artificial intelligence (AI) research.

Causes of AI Winter

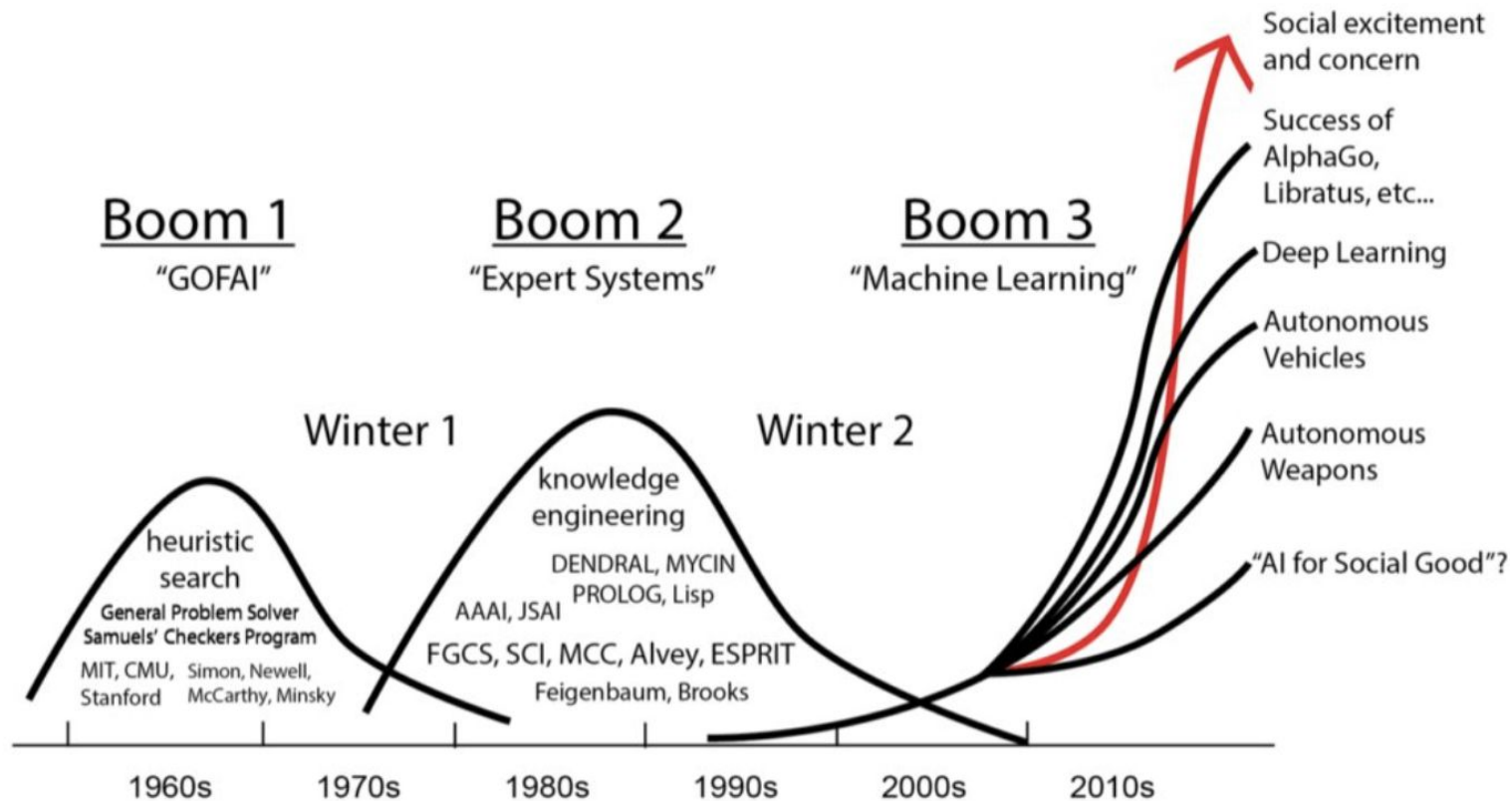
- **Unrealistic expectations:** When AI systems fail to live up to exaggerated claims
- **Lack of tangible results:** If AI research doesn't produce practical applications or significant breakthroughs
- **Economic downturns:** General economic recessions can reduce funding for all research areas, including AI.

AI Summers & AI Winters

Expectations



The Three Booms



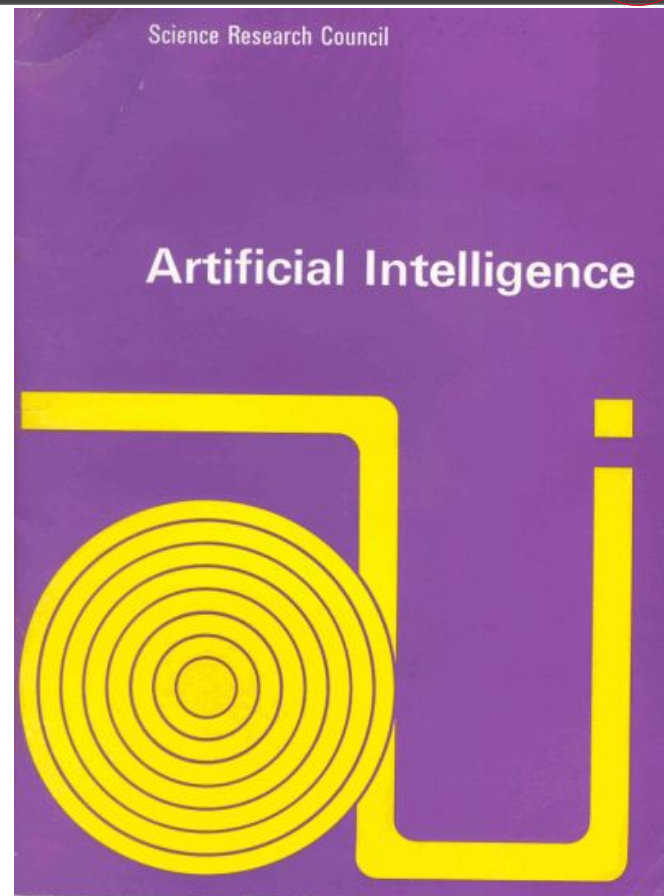
The Advent of the First AI Winter (1974-1980)



- 1969: Mansfield Amendment limited military funding for research that lacked a direct or apparent relationship to a specific military function.
- 1973: James Lighthill for British Science Research Council

“In no part of the field have discoveries made so far produced the major impact that was then promised.”

- DARPA withdrew its funding from many companies, including CMU’s \$3M of annual grants for speech.



Lighthill Report

Revival (1980s)



- 1980: Prof. McDermott writes XCON, an AI-powered expert system program saves Digital Equipment Corporation (DEC) 25 million dollars each year.
- 1981: The Japanese government invested hundreds of millions of dollars in projects aimed at making rapid leaps in AI.
- 1982: John Hopkins proved how a neural network could 'learn.' Geoffrey Hinton and David Rumelhart created backpropagation, reviving the field of connectionism.



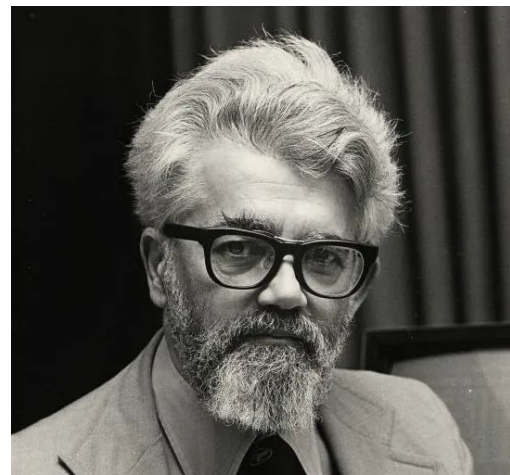
The Second AI Winter



1984: John McCarthy criticized expert systems - lack of common sense and their inability to understand their own limitations.

1987: Apple and IBM producing general purpose computers & solving more real-world problems ... much cheaper than any of the expensive AI-based systems.

Late 1980s: DARPA and Strategic Computing Initiative cut AI funding - did not trust the technology's capability to deliver results.



John McCarthy

DARPA Director Schwarz – "... very limited success in particular areas, followed immediately by failure to reach the broader goal at which these initial successes seem at first to hint..."

By 1991: Japan's Fifth Generation Computer project had finished 10 years, spent \$400 million, but hadn't met even one of the original expectations of the project.



- Statistical models for NLP analyses rose dramatically
- N-Grams are useful for clumps of linguistic data
- 1997 LSTM RNNs are introduced (later 2007 found their niche)

SRI International (now unaffiliated with Stanford)

- 2007, SRI spins off Siri, Inc., raises \$24M
- 2010, Apple acquires Siri, Inc.
- 2011, Siri becomes a feature in Apple iPhone 4S



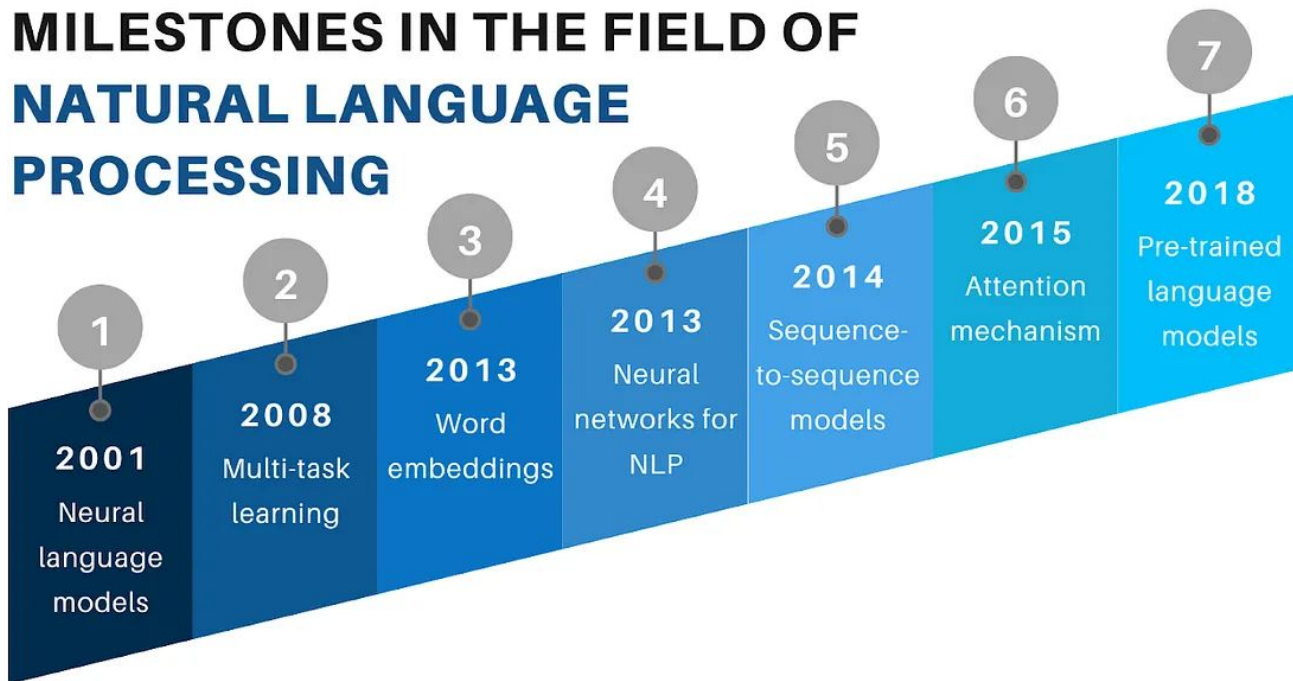
Section 3: Some History and Where it Pertains to You



- Prehistoric Natural Language Processing
- **Natural Language Processing: Third Boom**
- Applied NLP in the Age of LLMs



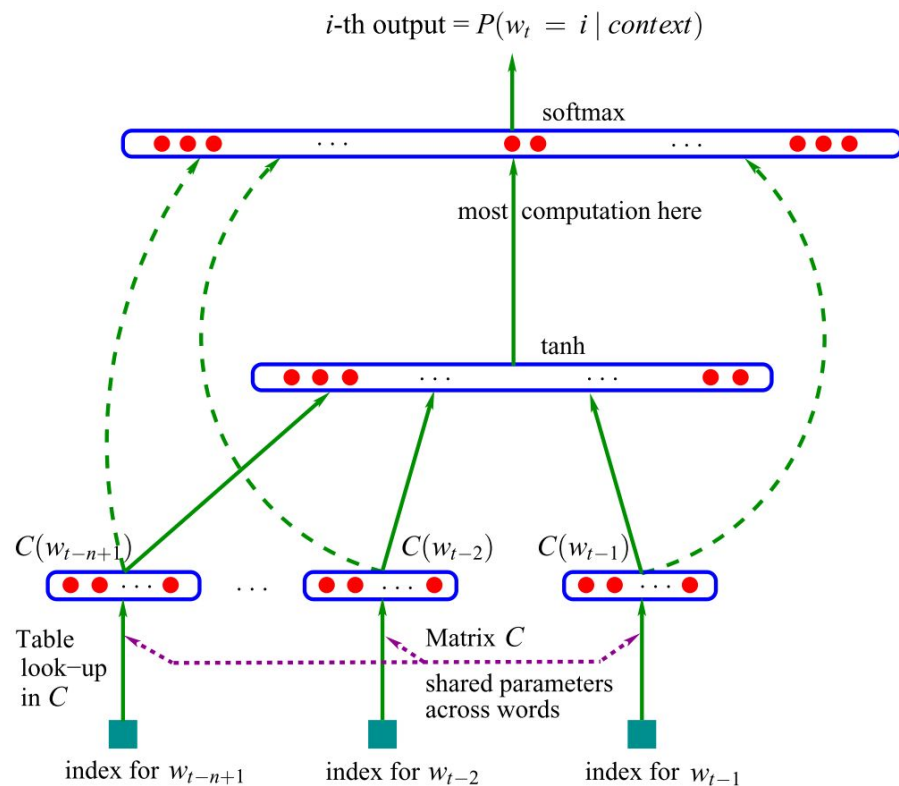
MILESTONES IN THE FIELD OF NATURAL LANGUAGE PROCESSING

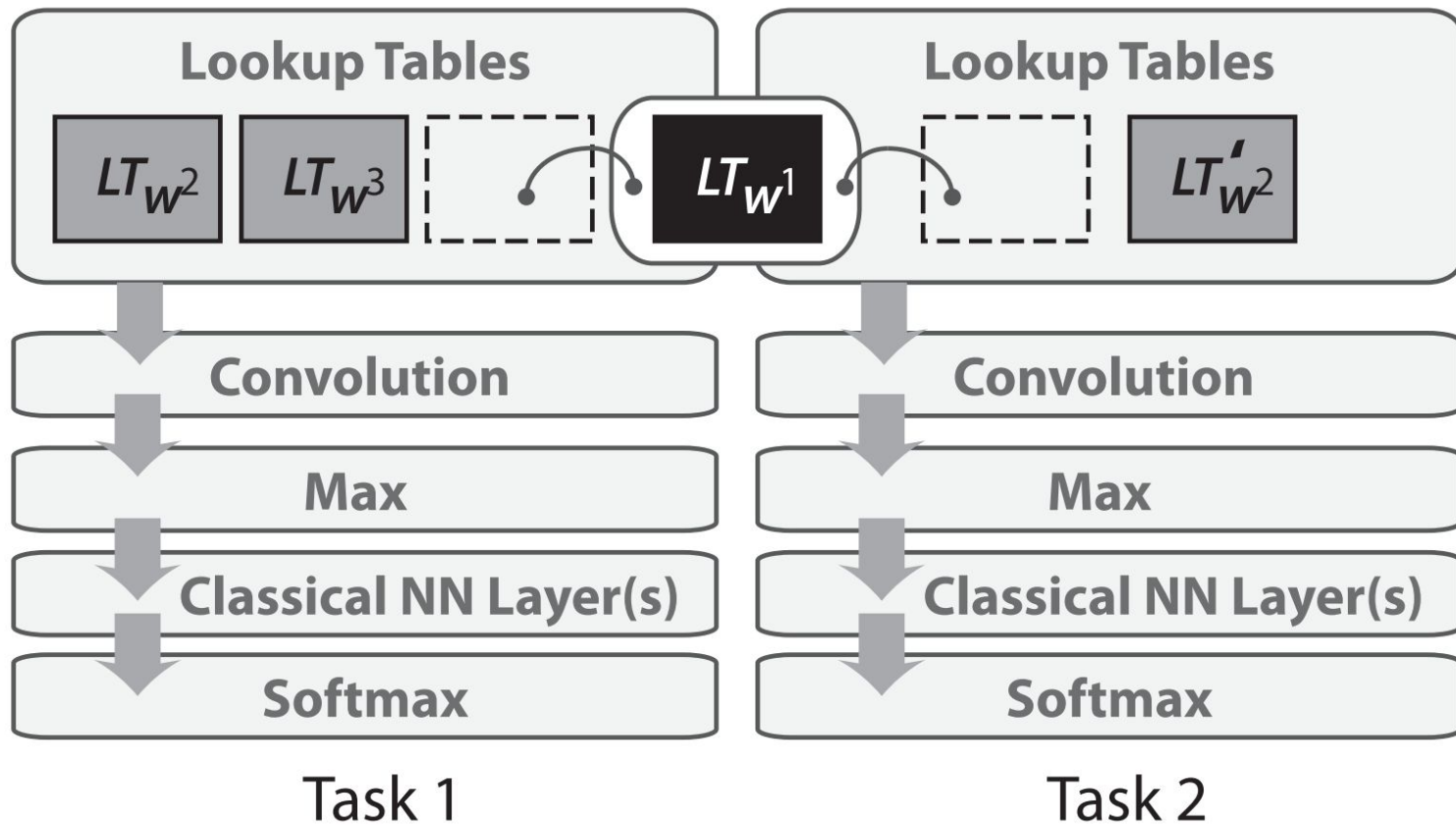


The First Neural “Language” Model (2001)

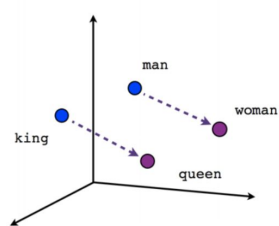


- Look up vector representations of n previous words (looked up in C)
- Embeddings concatenated and passed into DNN.
- Final output into prediction layer for the next word. .

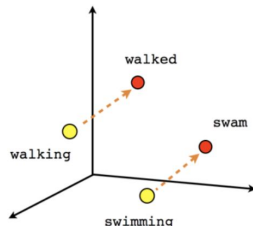




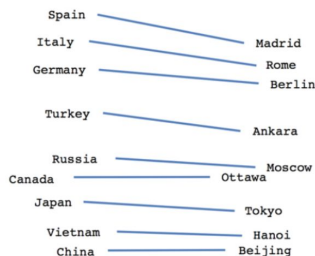
Word Embeddings (2013)



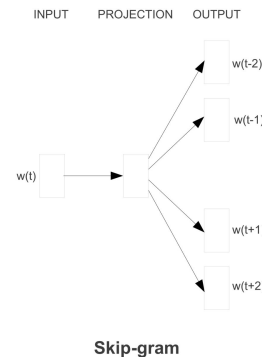
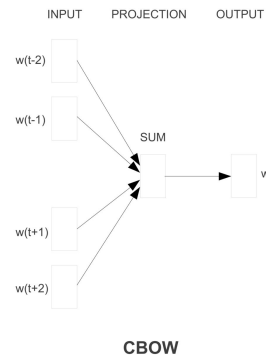
Male-Female



Verb tense



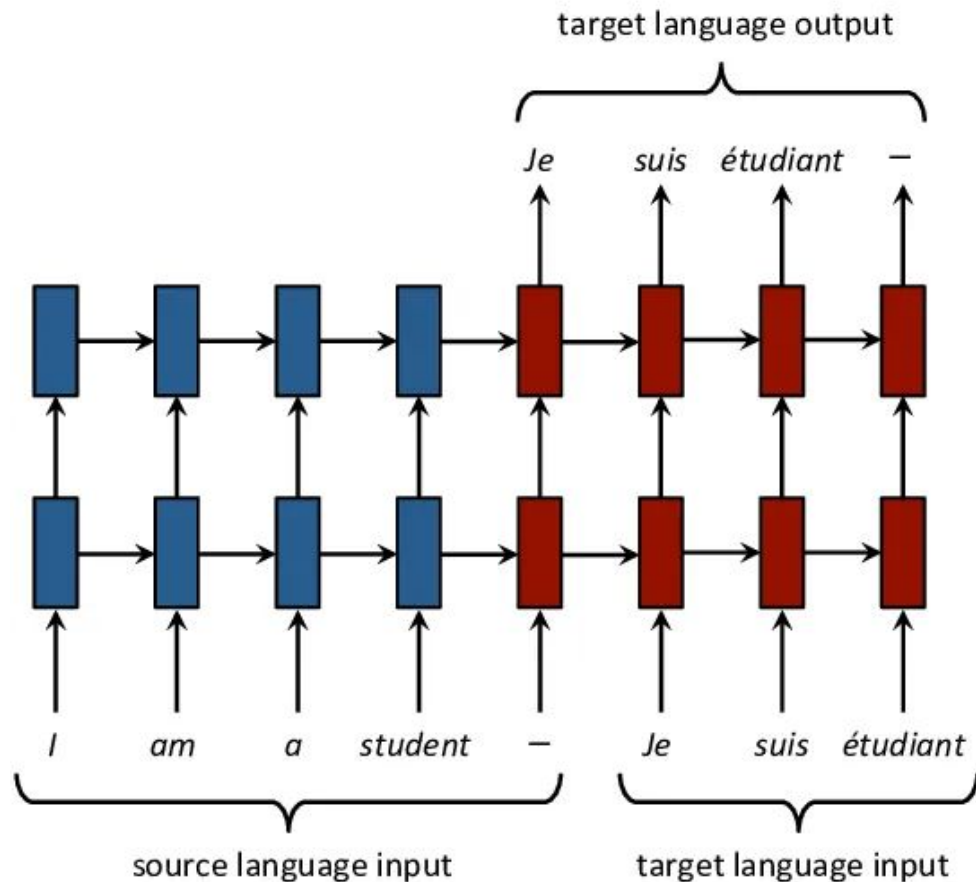
Country-Capital



- Deep conv neural networks for images (2012) - takes the world by storm
- **Google** - Mikolov creates word embeddings, terms it a DNN approach (2013)
- Derived technologies (i.e., negative sampling) find their way in optimization in modern information retrieval and other applications

Regains its footing for:

- Translation
- Speech to Text
- Most all longer-term NLP



Wholesale Investments in Neural Language Processing (2015)



A major drawback of statistical methods is that they require elaborate feature engineering. Since 2015,[22] the statistical approach was replaced by the neural networks approach, using semantic networks[23] and word embeddings to capture semantic properties of words.

The screenshot displays the Google Translate interface in two states. The top state shows the English input "The man shot the elephant while wearing his pyjamas." being translated into Hindi: "उस आदमी ने हाथी को अपना पजामा पहनाया।" (us aadamee ne haathee ko apana pajaama pahanaaya.). A red arrow points from the Hindi translation back to the English input. The bottom state shows the Hindi input "उस आदमी ने हाथी को अपना पजामा पहनाया।" being translated back into English: "The man put his pajamas on the elephant." This illustrates a significant semantic error where the original action of "shooting" is lost and replaced with "putting on".

Attention Modeling



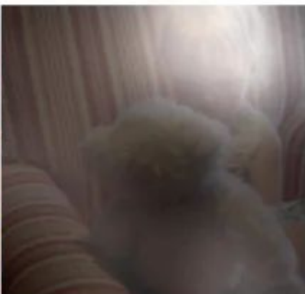
A woman is throwing a frisbee in a park.



A dog is standing on a hardwood floor.



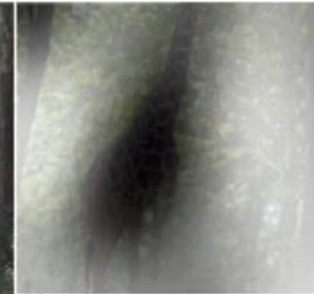
A stop sign is on a road with a mountain in the background.



A little girl sitting on a bed with a teddy bear.



A group of people sitting on a boat in the water.

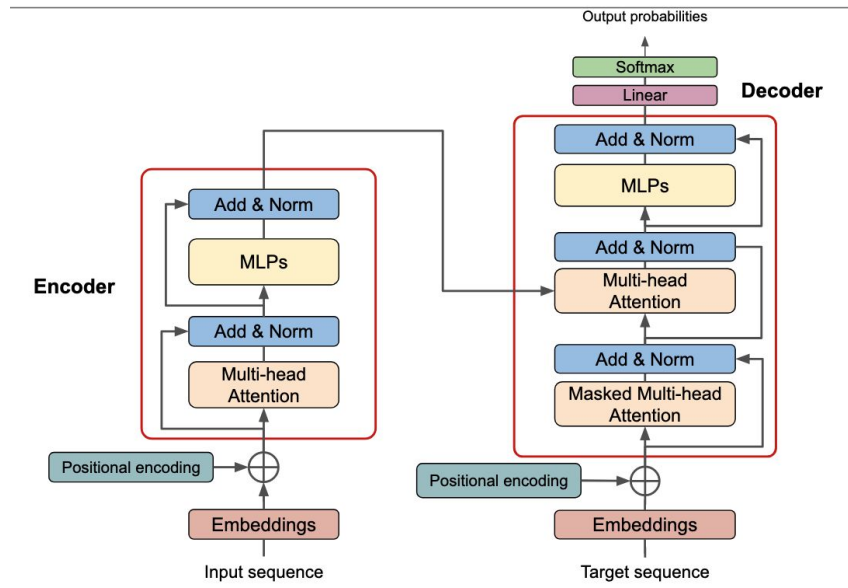


A giraffe standing in a forest with trees in the background.

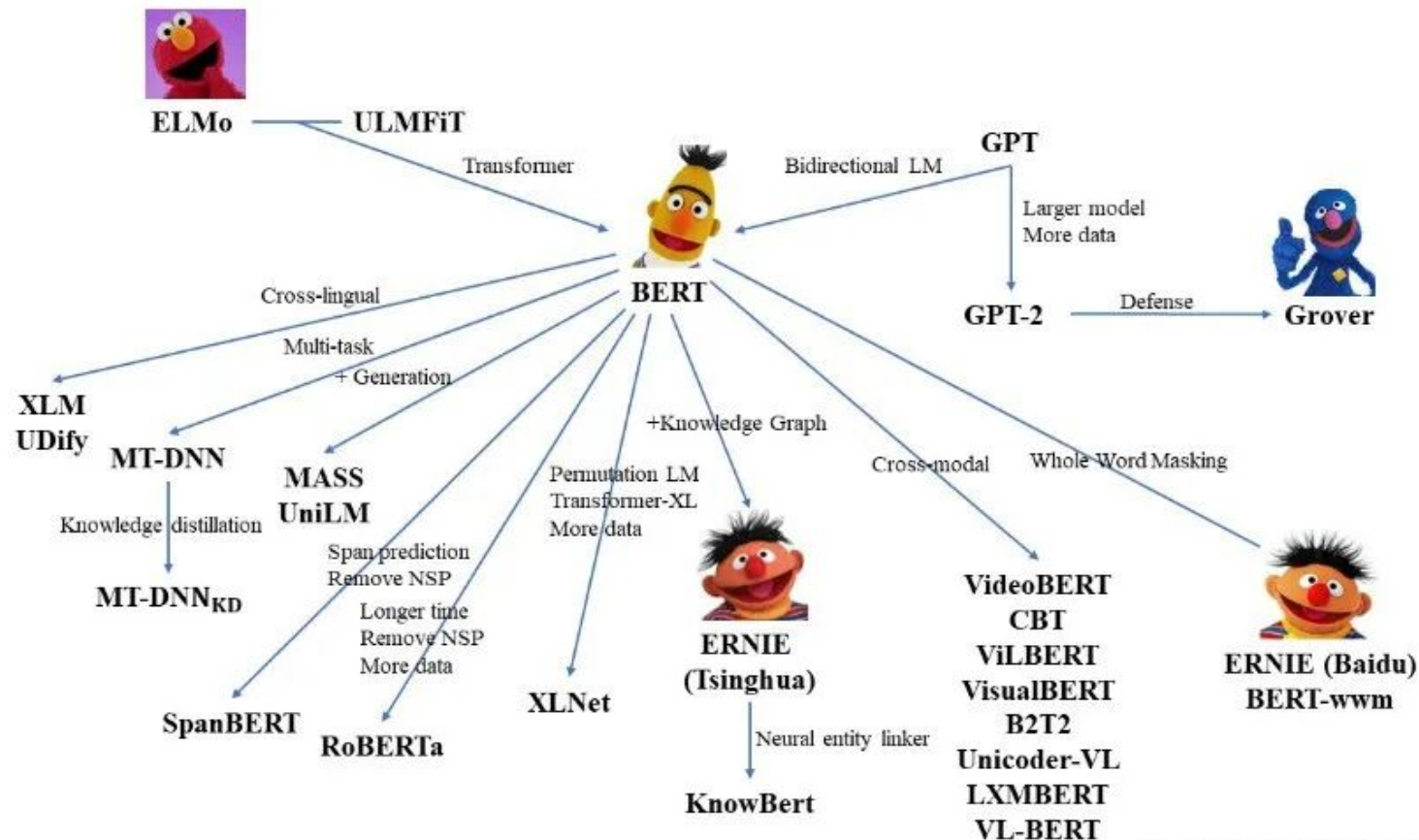
The Transformer (2017)



- Attention is All You Need
- First innovation that ***started*** with NLP modeling
- Google and the BERT model
- Almost every modern neural network model



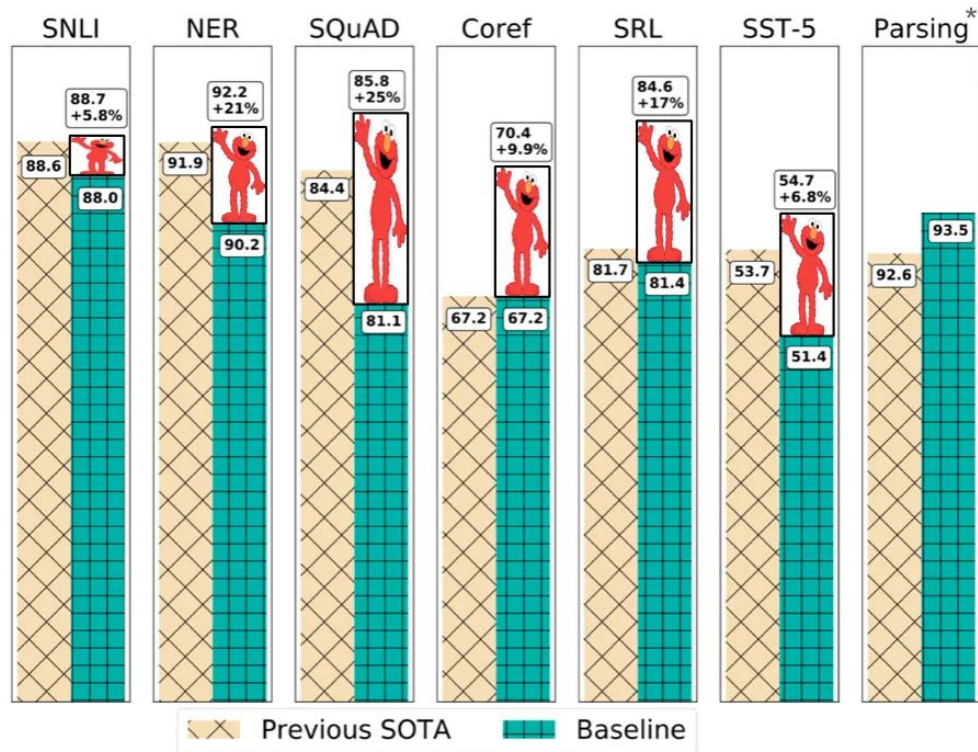
Transformer Modeling Engagements (2017+)



Pre-Trained Language Models



- Pretrained language models:
These methods use representations from language models for transfer learning.



*Kitaev and Klein, ACL 2018 (see also Joshi et al., ACL 2018)

Section 3: Some History and Where it Pertains to You

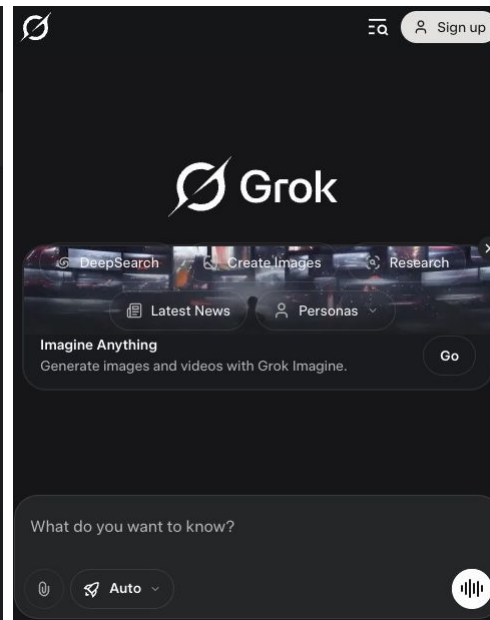
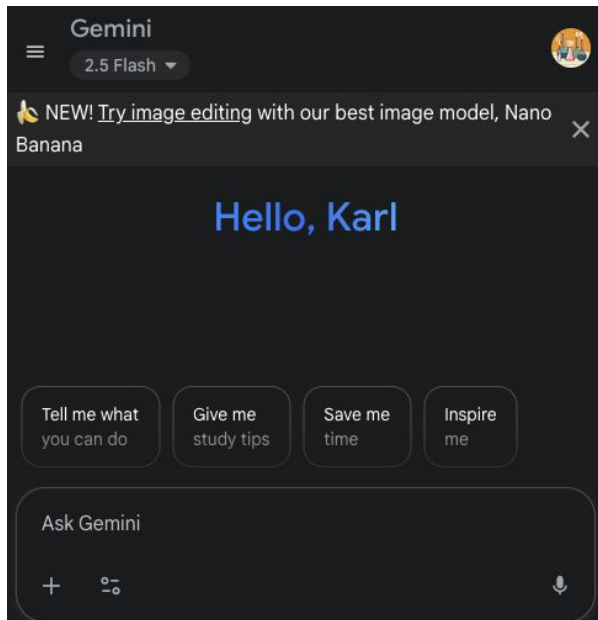
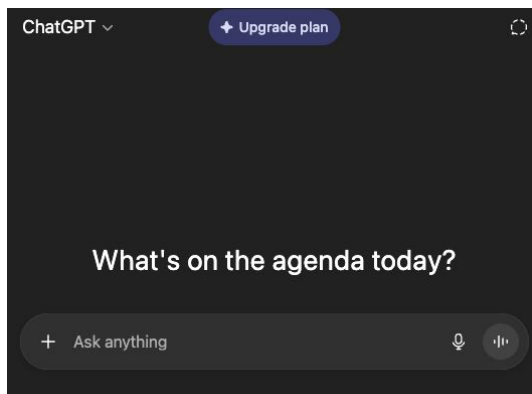
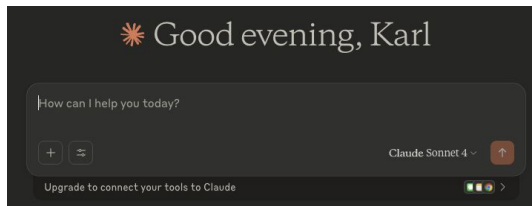


- Prehistoric Natural Language Processing
- Natural Language Processing: Third Boom
- **Applied NLP in the Age of LLMs**

Which ones do you use? For what?



- Who uses Cursor? How do you develop?
- Which LLMs do you use?



Large Models Reach Billions of Parameters



Largest BERT: 340M parameters.
(default / base = 110M) - 2021

GPT-2: 1.5B parameters

GPT-3: 340M parameters

GPT-4: 1.8T parameters

GPT-5: 17T parameters

```
from openai import OpenAI
client = OpenAI()

response = client.responses.create(
    model="gpt-5",
    input="Write a one-sentence bedtime story about a unicorn."
)

print(response.output_text)
```

OpenAI releases ChatGPT (2018)



- While OpenAI is founded in 2015, it becomes for-profit in 2019
- In 2022, OpenAI creates ChatGPT
- In Rapid Succession, the GPT 3.5, GPT 4, GPT 4o, GPT o1, GPT 5
- GPT-4o Advances and Multimodal Responses
- GPT-5 Latest but Most Advanced Reasoning Model



Proprietary Weights

- GPT-5
- Gemini 2.5 Flash
- Anthropic Claud 3 Family
- xAI Grok-4



Do you know of:

- AI21 Jamba Instruct
- Cohere Command R/R+
- Now LLM
- Titan

Open Weights

- GPT-OSS (120B, 20B)
- DeepSeek
- Qwen
- Mistral 3
- Llama 4

Do you know of:

- Databricks DBRX
- Snowflak Arctic
- Microsoft Phi-3
- Google Gemma 2

OpenAI's GPT models continue to lead the space, but today they face stiff competition—especially from Anthropic's Claude series, Google's Gemini family, and Microsoft/GitHub's Copilot (which now supports agentic functionality).

LLMs proliferate throughout the world



Beyond ChatGPT, Gemini, Claud, Llama in the United States:

- number of LLMs = 3,755 globally (as of Sept 2025)
- 1,509 are released from China, the largest single source by count
- China: pragmatic, low-cost scaling and broad deployment (lots of smaller/efficient models and open releases)
- U.S.: investment in fewer, top-end models and commercial APIs (OpenAI, Anthropic, Google)



Models Interact with External Applications



- Virtually all major LLM providers are agentic
- Foundation LLMs are “brain” → but architecture access external tools (including Search)

Feature	ChatGPT	Gemini	Claude	Copilot
Search Integration	Yes. Built-in search, previously with Bing.	Yes. Deep integration with Google Search.	Yes. Server-side "Tool use" for web browsing.	Yes. Deeply integrated into Edge browser.
Ecosystem Integration	Independent platform, but has custom GPTs.	Seamless with Google Workspace and Android.	Focus on API and documents via <code>claude.ai</code> .	Embedded in Microsoft 365, Windows, and Edge.
Primary Focus	General conversation and creative tasks.	Multimodal and Google ecosystem integration.	Safety, nuanced reasoning, long context analysis.	Work automation and context-aware browsing.
Visual/Multimodal	Strong image generation and vision capabilities.	Advanced visual reasoning across images and video.	Good with images, but excels at text and analysis.	Integrates with Edge for browsing, images via DALL-E.
Developer Tools	Robust API and customizable GPTs.	Strongest developer tooling for enterprise integration.	API-focused with new Artifacts feature.	Primarily an end-user product, but relies on Azure AI.



Production-level LLMs are evolving into agentic systems that interface deeply with external tools, services, and operational workflows.

- Agentic AI now autonomously orchestrates multi-step reasoning, planning, and external tool invocation, guided by human oversight ([Financial TimesTechRadar](#))
- Enterprise blueprints like the XDO framework (Experience–Data–Operations) focus on integrating such agentic models to automate tasks across IT and business processes ([TechRadar](#))
- Architectures now distinguish between fixed workflows (predictable) and more flexible, agent-driven sequences (dynamic)—highlighting this shift toward autonomy ([Diagrid](#))
- Tools exist to help agents securely access external systems and services—e.g., Anthropic’s Model Context Protocol (MCP)—a growing area of focus for production deployment

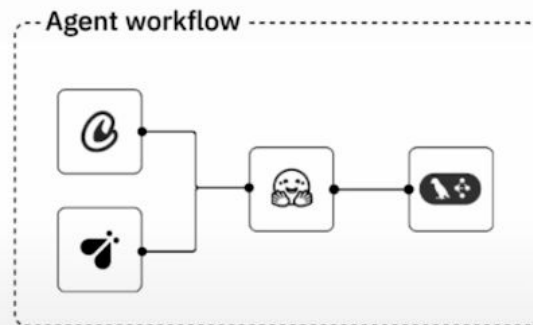
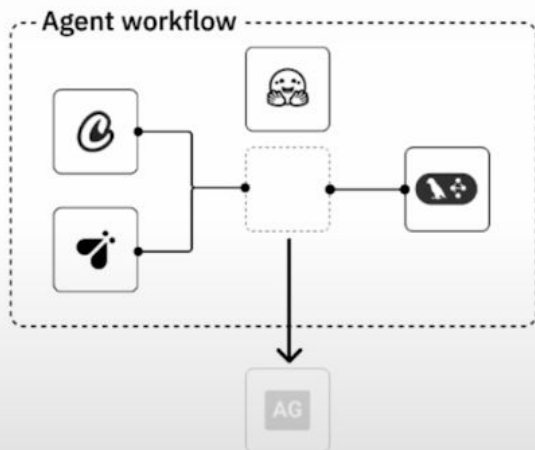
LLMs deployed in production increasingly act as agentic systems—capable of planning, tool use, and dynamic decision-making—enabling them to integrate into and orchestrate complex workflows

Connecting a Diverse Ecosystem of Stand Alone Agents



Agentic AI (Need for ACP)

Developer doesn't
need to write
integration logic



End user receives
better results

We're already seeing LLM agents interact with one another—forming collaborative, multi-agent ecosystems (sometimes referred to as the 'Agentic Web'), and frameworks like AgentNet, MetaGPT, and AutoGen.